



CNS2105: DATA AND VOICE COMMUNICATIONS: TUTORIAL QUESTIONS

1. Write down the range of frequencies of each of the following signals
 - a) Voice
 - b) Video
2. Distinguish between digital and analog signals
3. State three merits of modulation in communication systems
4. Draw a labelled diagram of a pulse code modulation (PCM) system and explain the function of each block.
5. A PCM system uses a uniform quantizer followed by a 7-bit encoder. The bit rate of the system is equal to 50×10^6 bits/second.
 - a) What is the maximum message bandwidth for which the system operates satisfactorily
 - b) Determine the output signal-to-(quantization) noise ratio when a full-load sinusoidal waveform of frequency 1MHz is applied to the input.
6. State the Nyquist sampling criterion
7. A speech signal has a total duration of 10s. It is sampled at the rate of 8kHz and then encoded. The signal to quantization noise ratio is required to be 40dB. Calculate the minimum storage capacity needed to accommodate this digitized speech signal.
8. We have sampled a low-pass signal with a bandwidth of 200Khz using 1024 levels of quantization.
 - a) Calculate the bit rate of the digitized signal
 - b) Calculate the SNR_{dB} of this signal

9. A carrier is modulated to a depth of 40%, find the percentage increase in the transmitted power (8 percent)
10. A certain AM transmitter is radiating 125kW when a certain audio sine wave is modulating it to a depth of 70 % and 144.5kW when a second sinusoidal wave also modulates it simultaneously. What is the depth of modulation for the second wave
11. What is the bandwidth of an AM modulated signal? What is the maximum transmission efficiency of a sinusoidal AM signal?
12. (a) Define each of the following as used in angle modulation
 - i. Phase modulation
 - ii. Frequency modulation
13. A voice signal and a carrier signal are each given by $m(t) = A_m \cos \omega_m t$ and $S_c(t) = A \cos \omega_c t$. Write down the expressions for the :
 - a) AM modulated waveform
 - b) Frequency modulated waveform
 - c) Phase modulated waveform
14. With the aid of a block diagram, show how to obtain
 - a) A FM modulator from a PM modulator
 - b) A PM modulator from a FM modulator
15. Show how the following bit rates are arrived at in voice transmission standards hierarchy.
 - a) E1 system rate of 2.048Mbps
 - b) T1 system rate of 1.544 Mbps
16. A transmitter radiates a DSB-AM signal with a total power of 5Kw at a depth of modulation of 60 %. Calculate the power transmitted in the carrier and also in each sideband.

[4.24Kw 0.38Kw]

17. A carrier wave represented by $10 \cos(2\pi 10^6 t) V$ is amplitude modulated by a second wave represented by $3 \cos(2\pi 10^3 t) V$. Calculate

- a) The depth of modulation
- b) The upper and lower side frequencies
- c) The amplitude of the side frequencies
- d) The fraction of the power transmitted in the sidebands

[(a)30% (b)1.001MHZ, 0.999MHZ (c) 1.5V (d)4.3%]

18. A DSB-AM transmitter produces a total output of 24kW when modulated to a depth of 100%. Determine the power output when

- a) The carrier is unmodulated
- b) The carrier is modulated to a depth of 100%, one sideband is suppressed and the carrier component is attenuated by 26dB

[(a)16kW (b)1.48 kW]

19. A frequency modulated wave has a total bandwidth of 165kHz when the modulating signal is a single tone frequency of 10kHz. Find the maximum carrier deviation produced by the modulating signal.

Line Encoding schemes

- 1. Distinguish between data rate and modulation rate
- 2. Explain the following desirable properties of a line encoding scheme
 - a) Self-synchronizing
 - b) Transparency
 - c) Error correction capability
- 3. A data sequence 1100001111000010100 is to be encoded using a line code. On the same axis, draw the waveforms for each of the following line encoding schemes:

- a) Manchester
 - b) Differential Manchester
 - c) Bipolar AMI
 - d) HDB3
4. Explain the reason for scrambling of data sequence before encoding in data encoding